Task A: Differentiation of Objective Functions

Exercise A.1

Find the global minimum of f and 'show' that it is indeed a true minimum using the sign of the 2nd derivative at your candidate solution.

$$f(x) = (x-5)^2$$

Solution:

first derivative

$f'(x) = 2(x-5)$, Let's set $f'(x) = 0$

$\Rightarrow 2(x-5) = 0$

$x^* = 5$

second derivative

$f''(x) = 2$

then evaluate f'' at $x^* = 5$

$f''(5) = 2$ which is greater than 0.

Since $f''(5)$ is positive, $x = 5$ is a local minimum.

And, because $(x-5)^2 \geq 0$ for all x and $f(5) = 0$, $\underline{x^* = 5}$ is the global minimum.

Exercise A.2

find the global minimum of f and "show" that it is indeed a true minimum.

$$f(x) = -e^{-(x-2)^2}$$

Solution:

first derivative

$f'(x) = -\frac{d}{dx}\left(e^{-(x-2)^2}\right) = -e^{-(x-2)^2} \cdot (-2(x-2))$

$= 2(x-2)e^{-(x-2)^2}$, Let's set $f'(x) = 0$

$f'(x) = 0$

$= 2(x-2)e^{-(x-2)^2} = 0$

$\Rightarrow (x-2) = 0$

$\underline{x^* = 2}$; the exponential factor $(e^{-(x-2)^2})$ never zero.

Second derivative

$f''(x) = \frac{d}{dx}\left[2(x-2)e^{-(x-2)^2}\right] = 2e^{-(x-2)^2} + 2(x-2) \cdot e^{-(x-2)^2} \cdot (-2(x-2))$

$= 2e^{-(x-2)^2}(1 - 2(x-2)^2)$

Evaluate f'' at $x = 2$, $f''(2) = 2 \cdot e^0(1-0) = \underline{2 > 0}$

since the second derivative is positive, $\underline{x^* = 2}$ is a local minimum

* We can say that for any real x, $e^{-(x-2)^2}$ satisfies $0 < e^{-(x-2)^2} \leq 1$

Hence, $-1 \leq -e^{-(x-2)^2} < 0$; the smallest possible value of $f(x)$ is -1, and it can be achieved only when $e^{-(x-2)^2} = 1$, which means when $x = 2$

Combine w. the second derivative test showing a local minimum at $x = 2$, we can conclude that, the global minimum is at $\underline{x^* = 2}$ with $f(2) = -1$

Exercise 1.3

Find the global minimum of f and "show" that it is indeed a true minimum by means of the Hessian.

$$f(x,y) = 3.(x-2)^2 + 4.(y-1)^2$$

Solution:

first order partial derivative

$\frac{\partial f}{\partial x} = 6(x-2)$ and $\frac{\partial f}{\partial y} = 8(y-1)$, set both equal to zero.

$$\frac{\partial f}{\partial x} = 0 \Rightarrow 6(x-2) = 0$$

$$x = 2$$

$$\frac{\partial f}{\partial y} = 0 \Rightarrow 8(y-1) = 0$$

$$y = 1$$

Therefore, $f(2,1) = 0$

second order partial derivative

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x^2}(6(x-2)) = 6$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial y}(8(y-1)) = 8$$

$$\frac{\partial^2 f}{\partial x \partial y} = 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial x} = 0$$

The Hessian matrix looks like this: $\begin{bmatrix} 6 & 0 \\ 0 & 8 \end{bmatrix}$, w/c has eigen values $6, 8 > 0$

Since, the eigen values are positive definite, it shows that it's strict local minimum,

And, we know that $f \geq 0$ and $f(2,1) = 0$, w/c means

f at $(x^*, y^*) = (2,1)$ is global minimum $\bar{w}$ $f = 0$.

Exercise A.4

Determine the gradient of the function f at the point $(x_0, y_0) = (1, 3)$

$$f(x,y) = (y-2)^2 (x-5)^2 + (x-5)^2 + (y-2)^2$$

Solution :

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left((y-2)^2 (x-5)^2 + (x-5)^2\right)$$

$$= (y-2)^2 \, 2(x-5) + 2(x-5)$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left((y-2)^2 (x-5)^2 + (y-2)^2\right)$$

$$= 2(y-2)(x-5)^2 + 2(y-2)$$

Now, let's compute the gradient at $(x_0, y_0) = (1, 3)$

$$\left.\frac{\partial f}{\partial x}\right|_{(1,3)} = (3-2)^2 \cdot 2(1-5) + 2(1-5)$$

$$= 1^2 \cdot 2(-4) + 2(-4)$$

$$= \underline{\underline{-16}}$$

$$\left.\frac{\partial f}{\partial y}\right|_{(1,3)} = 2(3-2)(1-5)^2 + 2(3-2)$$

$$= 2 \cdot (1)(-4)^2 + 2(1)$$

$$= 2 * 16 + 2$$

$$= \underline{\underline{34}}$$

Therefore, Gradient at $(1,3)$; $\nabla f(1,3) = \begin{bmatrix} -16 \\ 34 \end{bmatrix}$

Exercise 1.5

Try to find the global minimum of f:

$$f(x,y) = (y-2)^2 (x-5)^2 + (x-5)^2 + (y-2)^2$$

Solution:

The function f combines quadratic terms in both x & y, so it's always nonnegative.
The only way to make f as small as possible is to set $x-5=0$ and $y-2=0$
so candidate point (x,y) is $(5,2)$

$f(5,2)=0$, since $f \geq 0$ everywhere; there are no negative values because each term is squared.

Hence, zero (0) is global minimum value at $(x^*,y^*) = (5,2)$

We can verify this with Hessian Matrix

We have found the first derivate in the exercise 1.4

$$\frac{\partial f}{\partial x} = (y-2)^2 . 2(x-5) + 2(x-5)$$

$$\frac{\partial f}{\partial y} = 2(y-2) . (x-5)^2 + 2(y-2)$$

set $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ equal to 0.

and we find $x=5$ and $y=2$

$(x^*,y^*) = (5,2)$, $f(x^*,y^*) = 0$

Second partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(2(x-5)\left((y-2)^2+1\right)\right)$$

$$= 2\left((y-2)^2+1\right)$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(2(y-2)\left((x-5)^2+1\right)\right)$$

$$= 2\left((x-5)^2+1\right)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}\left(2(x-5)\left((y-2)^2+1\right)\right)$$

$$= 2(x-5) . 2(y-2)$$

$$= 4(x-5)(y-2)$$

$$\frac{\partial^2 f}{\partial y \partial x} = 4(x-5)(y-2)$$

, no we can avaluate at the candidate point $(x^*,y^*) = (5,2)$

The Hessian matrix becomes $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, the eigenvalues are 2 & 2, which are positive.

* Positive definite Hessian at a stationary point implies a strict local minimum.
* Because $f \geq 0$ for all (x,y) and $f(5,2)=0$, this local minimum is also the global minimum.

Excercise 8.1

The sise of the current step is determined by the user defined parameter η, which often is reffered to a "step size" or "step lenght". However, this terminology is actually misleading: Explain the fact that the true step size is not η_t but instead $\eta_t \|\nabla f(x_t)\|$.

Solution:

The gradient descent update rule

$$x_{t+1} = x_t - \eta_t \nabla f(x_t)$$

The true step size is the actual distance we move from one iteration to the next

$\therefore$ step size $= \|x_{t+1} - x_t\|$, let's substitute this equation in the update rule equation

which is, $x_{t+1} = x_t - \eta_t \nabla f(x)$

$x_{t+1} - x_t = -\eta_t \nabla f(x)$, let's put both side of equation in the absolute value

$$\|x_{t+1} - x_t\| = \|\eta_t \nabla f(x)\|$$

$$\|x_{t+1} - x_t\| = \eta_t \|\nabla f(x)\|$$

So, the true step size (the actual distance we move) depends on both:

- η_t is a scalar or scaling factor, not the distance itself.
- the gradient magnitude $\|\nabla f(x_t)\|$; the actual distance moved (step size) changes every iteration because the gradient changes.

Exercise 8.2 - Stoping Criteria

Solution:

When we run Gradient Descent, we need a way to decide when to stop the iteration like the one we use, the base case, for recursive function. There are multiple stoping criteria, and each has pros and cons.

1 st criterion: Maximum no of iterations

$$\text{if } t \geq t_{max} \text{ , break the iteration}$$

- ~~It~~ stop ~~when the gradient becomes very small~~ after a fixed number of steps (e.g. 1000 iterations), regardless of whether convergence is reached.

pros:
- prevents infinite loops
- simple and always works as a safety measure.

cons:
- Doesn't guarantee convergence; it might stop too early or too late.
- Gives no information about whether the algorithm found minimum.

In general, we can always include this criterion as a safety measure.

2^{nd} criterion: Small gradient magnitude

$$\|\nabla f(x_t)\| < T$$

- ~~It~~ stops when the slope (gradient) becomes very small; meaning the algorithm is near a stationary point (flat region).

pros:
- It's theoretically sound; when the gradient is near zero, ~~we're~~ close to an optimum.
- Indicate true convergence.

cons:
- If the surface is flat or noisy, gradient might ~~be~~ small even when not at the global minimum.
- Can get stuck in local minima or saddle points.

* It helps detect convergence properly.

3^{rd} criterion: Small change in function value

$$|f(x_{t+1}) - f(x_t)| < \epsilon$$

- It stops when the improvement in the objective function becomes negligible.

pros:
- ~~It~~ stops when further progress doesn't improve the result meaningfully.
- Works well when we care about the final function value, not just position.

cons:
- Sensitive to scalling of f(x); may need to tune ϵ.
- Doesn't catch oscillation or divergence.

4th Criterion: Small change in parameter values

$$\|x_{t+1} - x_t\| < \delta$$

It stops when updates to x are very small; it's barely moving anymore.

Pros:
- Indicates stabilization of parameters
- Easy to compute

Cons:
- Might falsely stop if the learning rate or step length η is too small
- Doesn't guarantee we're at minimum (may be just slowed down.